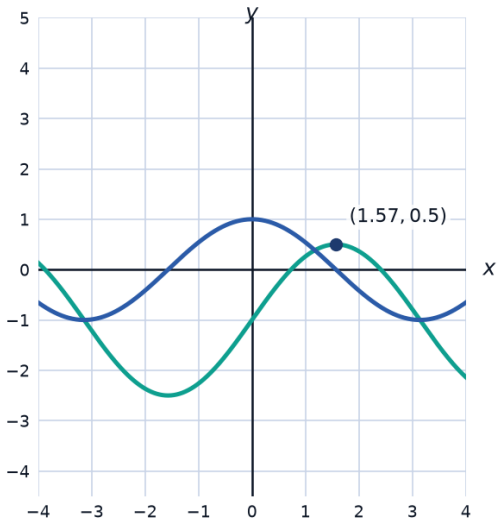
Trigonometry & Polar Coordinates

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

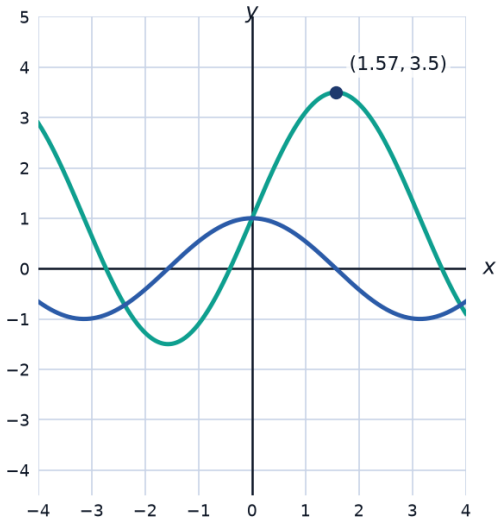
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Convert the polar coordinates $\left(6, \frac{5\pi}{4}\right)$ into rectangular Cartesian coordinates (x, y).



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Convert the Cartesian equation $x^2 + (y - 4)^2 = 16$ into a polar equation relating r and θ.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Evaluate the trigonometric limit: $\lim_{x \to 0} \frac{\sin(5x)}{\tan(2x)}$

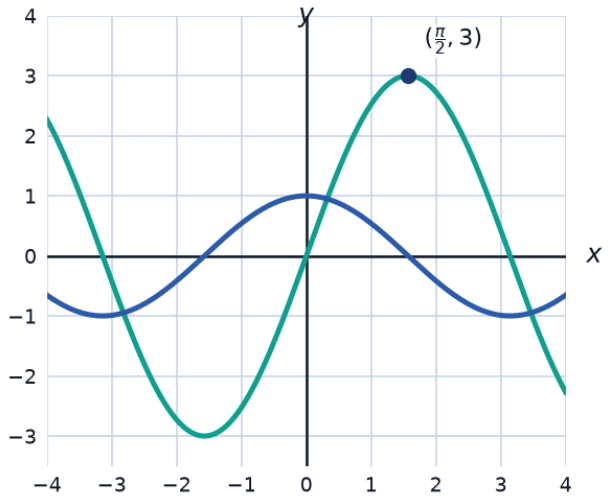
4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the trigonometric equation on the interval $[0, 2\pi)$: $2\cos^2 x - 3\cos x + 1 = 0$

© ClipSAT — clipsat.org | All Rights Reserved

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Evaluate the limit: $\lim_{x \to 0} \frac{\sqrt{1+x}-1}{x}$



6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the scalar projection of $\mathbf{u} = \langle 4, 3 \rangle$ onto $\mathbf{v} = \langle 12, 5 \rangle$.

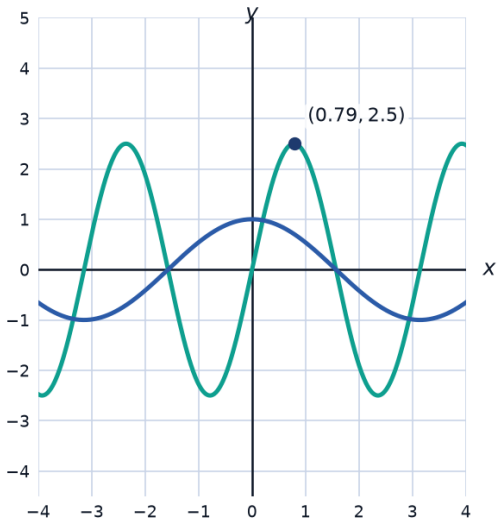
7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the power of the imaginary unit: $i^{2026} + i^{2027} + i^{2028} + i^{2029}$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A constant force vector $\mathbf{F} = \langle 8, -3 \rangle$ (in newtons) moves an object along the displacement vector $\mathbf{d} = \langle 5, 2 \rangle$ (in meters). Calculate the work done by the force.



© ClipSAT — clipsat.org | All Rights Reserved

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given that $\sin\theta = \frac{3}{5}$ with θ in Quadrant II, find the exact value of $\sin(2\theta)$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the indeterminate limit: $\lim_{x \to 5} \frac{x^2 - 25}{x^2 - 8x + 15}$

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

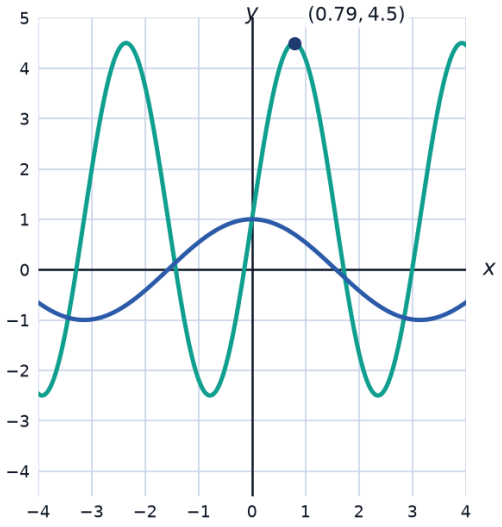
In triangle ABC, $A = 45^\circ$, $B = 60^\circ$, and $a = 8$. Find the exact length of side b.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Vectors $\mathbf{u} = \langle 3, -2, 5 \rangle$ and $\mathbf{v} = \langle 4, k, -2 \rangle$ are orthogonal in $\mathbb{R}^3$. Find the value of k.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Evaluate the limit at infinity: $\lim_{x \to \infty} \frac{5x^3 - 2x + 7}{2x^3 + 4x^2 - 1}$



14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Given $A = (2, -1; 3, 4)$ and $B = (0, 5; -2, 1)$, compute $3A - 2B$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Advanced Functions & Equations

Use the Remainder Theorem to find the remainder when $P(x) = 3x^4 - 2x^3 + 5x - 7$ is divided by $x + 2$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Limits

In triangle ABC, $a = 7$, $b = 5$, and $\cos C = \frac{1}{5}$. (a) Find the exact length of side c. (b) Calculate the exact value of $\sin^2 C$ and the area of triangle ABC.

© ClipSAT — clipsat.org | All Rights Reserved

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Complex Numbers

Convert the polar equation $r = 6\cos\theta$ into Cartesian coordinates, identify the conic section, and state its center and radius.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Matrices & Vectors

A student spends 90 seconds on each of the first 20 questions of the 50-question AIST Math 2 exam. How much average time (in seconds) remains per question for the remaining 30 questions?

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Sequences & Series

Compute $(\sqrt{3} - i)^8$ using De Moivre's Theorem, expressing the result in standard rectangular form $a + bi$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Advanced Functions & Equations

Evaluate the limit at infinity involving a square root in the numerator: $\lim_{x\to\infty} \frac{\sqrt{9x^2 + 4x}}{2x - 1}$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express $2\sin(5x)\cos(3x)$ as a sum of trigonometric functions.

a. $\sin(8x) + \sin(2x)$

b. $\sin(8x) - \sin(2x)$

c. $\cos(8x) + \cos(2x)$

d. $\cos(8x) - \cos(2x)$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Simplify the expression $\frac{\sin(3x) + \sin(x)}{\cos(3x) + \cos(x)}$.

a. $\tan(2x)$

b. $\tan(x)$

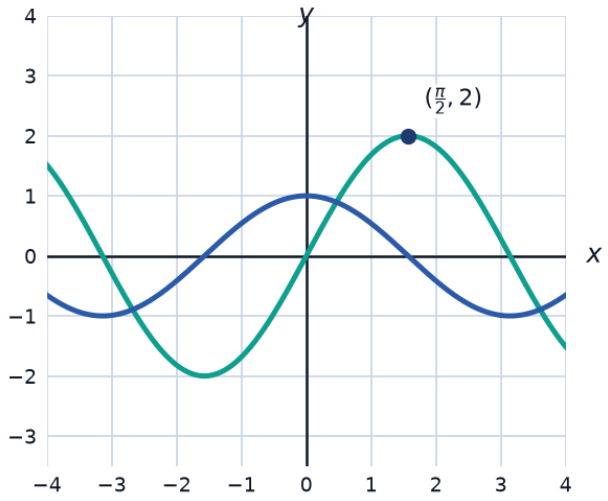
c. $\cot(2x)$

d. $\tan(4x)$

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the exact value of $\cos(105^\circ) - \cos(15^\circ)$?



a. $-\frac{\sqrt{2}}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $-\frac{\sqrt{6}}{2}$

d. $\frac{\sqrt{6}}{2}$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What geometric curve is represented by the polar equation $r = 2 + 2\cos\theta$?

a. Cardioid

b. Limaçon with inner loop

c. Three-petal rose

d. Circle

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many petals does the rose curve $r = 4\cos(3\theta)$ have?

a. 3

b. 6

c. 4

d. 8

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the polar point $(4, \frac{5\pi}{6})$ to rectangular coordinates (x, y).

a. $(-2\sqrt{3}, 2)$

b. $(2\sqrt{3}, -2)$

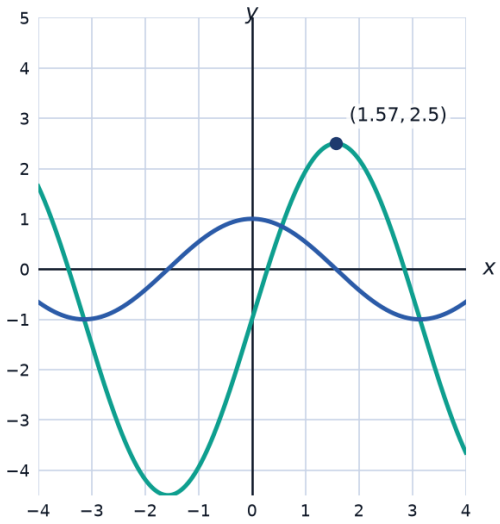
c. $(-2, 2\sqrt{3})$

d. $(2, -2\sqrt{3})$

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the rectangular point $(-3, -3)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.



a. $(3\sqrt{2}, \frac{5\pi}{4})$

b. $(3\sqrt{2}, \frac{3\pi}{4})$

c. $(6, \frac{5\pi}{4})$

d. $(3\sqrt{2}, \frac{7\pi}{4})$

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the Cartesian equation for the polar line $\theta = \frac{\pi}{3}$?

a. $y = \sqrt{3}x$

b. $y = \frac{\sqrt{3}}{3}x$

c. $y = x + \frac{\pi}{3}$

d. $x = \sqrt{3}y$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find all solutions to $\sin(2x) = \cos(x)$ in the interval $[0, 2\pi)$.

a. $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$

b. $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$

c. $\frac{\pi}{3}, \frac{2\pi}{3}$

d. $\frac{\pi}{6}, \frac{5\pi}{6}$

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the range of $f(x) = 2 - 3\sin(4x)$?

a. $[-1, 5]$

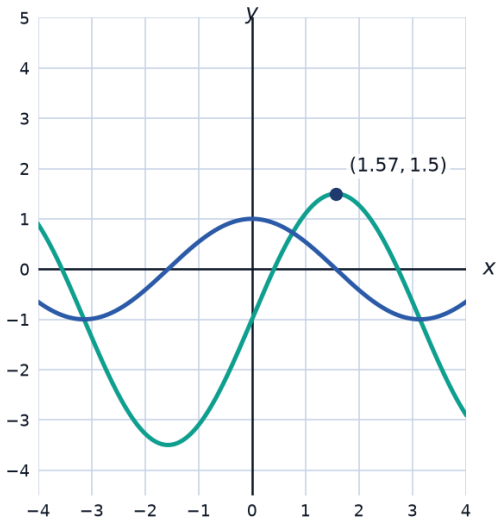
b. $[-3, 3]$

c. $[-5, 1]$

d. $[2, 5]$

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the standard (x, y)-coordinate plane, what are the rectangular coordinates of the point with polar coordinates $(r, \theta) = \left(6, \frac{5\pi}{6}\right)$?



a. $(-3, 3\sqrt{3})$

b. $(-3\sqrt{3}, 3)$

c. $(3\sqrt{3}, -3)$

d. $(-3\sqrt{3}, -3)$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A point in the polar coordinate plane has coordinates $(r, \theta) = \left(4, \frac{5\pi}{6}\right)$. Which of the following gives the rectangular coordinates (x, y) of this point?

a. $(-2\sqrt{3}, 2)$

b. $(-2, 2\sqrt{3})$

c. $(2, -2\sqrt{3})$

d. $(2\sqrt{3}, -2)$

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Convert the rectangular point $(-3.464, -2)$ (approximately) to polar form (r, θ), $0^\circ \le \theta < 360^\circ$.

a. $(4, 210^\circ)$

b. $(4, 30^\circ)$

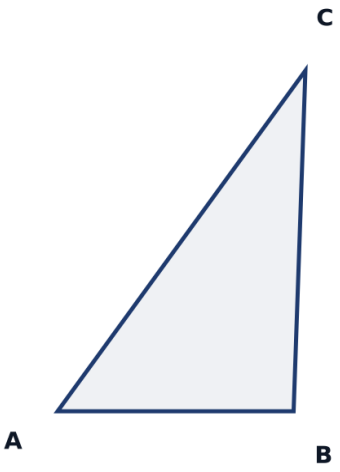
c. $(4, 150^\circ)$

d. $(8, 210^\circ)$

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC, $A = 54^\circ$, $a = 10$, $B = 92^\circ$. Find b to the nearest tenth.



a. 12.4

b. 13.4

c. 11.9

d. 10

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In the polar coordinate plane, a curve is defined by the equation $r = 6\cos\theta + 8\sin\theta$. Two rays emanating from the pole at angles $\theta = \alpha$ and $\theta = \alpha + \frac{\pi}{2}$ intersect the curve at points $PeqO$ and $QeqO$, respectively, where O denotes the pole. Assuming α is chosen such that both P and Q have positive radial coordinates, what is the maximum possible area of ΔOPQ?

a. 25

b. 24

c. 48

d. 50

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In the polar coordinate plane, points A and B lie on the curve $r = 4\sin(2\theta)$ such that their polar angles satisfy $0 < \theta_A < \theta_B < \frac{\pi}{2}$ and $\theta_B - \theta_A = \frac{\pi}{3}$. What is the maximum possible area of ΔOAB, where O is the origin?

a. $\sqrt{3}$

b. 2

c. $2\sqrt{3}$

d. $4\sqrt{3}$

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the polar coordinate system, point A has polar coordinates (r_A, θ_A) lying on the curve $r = 6\sin\theta$, and point B has polar coordinates (r_B, θ_B) lying on the curve $r = 8\cos\theta$, with $r_A > 0$ and $r_B > 0$. If $\theta_A - \theta_B = \frac{\pi}{6}$, what is the maximum possible area of ΔOAB, where O is the pole (origin)?

a. 6

b. $6\sqrt{3}$

c. 9

d. 12

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Matrices & Vectors

A point has polar coordinates $(r, \theta) = \left(6, \frac{2\pi}{3}\right)$. What are the Cartesian coordinates (x, y) of this point?

a. $(-3, 3\sqrt{3})$

b. $(-3\sqrt{3}, 3)$

c. $(3, -3\sqrt{3})$

d. $(3\sqrt{3}, -3)$

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Sequences & Series

A point in the polar coordinate system has coordinates $(r, \theta) = \left(6, \frac{5\pi}{6}\right)$. Which of the following represents the Cartesian coordinates (x, y) of this point?

a. $(-3\sqrt{3}, 3)$

b. $(-3, 3\sqrt{3})$

c. $(3\sqrt{3}, -3)$

d. $(-3\sqrt{3}, -3)$

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Advanced Functions & Equations

If $\sin x = \frac{7}{25}$ and $\cos x = \frac{24}{25}$ (Quadrant I), find $\sin(2x)$.

a. $\frac{336}{625}$

b. $\frac{527}{625}$

c. $\frac{7}{25}$

d. $\frac{24}{25}$

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Limits

In which quadrant does the terminal side of 65° lie?

a. Quadrant 1

b. Quadrant 2

c. Quadrant 3

d. Quadrant 4

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Complex Numbers

Convert $(r, \theta) = (2, 30^\circ)$ to rectangular coordinates.

a. $(3,\ 1)$

b. $(1,\ 3)$

c. $(-3,\ 1)$

d. $(3,\ -1)$

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 2.01 Matrices & Vectors

In the xy-plane, let $P(t, t^2)$ with $t > 0$ be a point on the parabola $y = x^2$. The normal line to the parabola at P intersects the parabola again at a second point Q. Let T be the intersection point of the tangent lines to the parabola at P and Q. As $t \to 0^+$, what is the limiting value of the y-coordinate of T?

a. $-\frac{1}{2}$

b. $-\frac{1}{4}$

c. -1

d. 0

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Sequences & Series

For real constants a and b, suppose that $\lim_{x \to 0} \frac{\sqrt{ax + b} - 3}{x} = 2$. What is the value of $a + b$?

a. 12

b. 15

c. 21

d. 27

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Advanced Functions & Equations

For each real number $t \in (-2, 2)$, the cubic equation $x^3 - 3x + t = 0$ has three distinct real roots, denoted in increasing order by $x_1(t) < x_2(t) < x_3(t)$. What is the value of the limit $\lim_{t \to 2^-} \frac{x_3(t) - x_2(t)}{\sqrt{2 - t}}$?

a. $\frac{2\sqrt{3}}{3}$

b. $\sqrt{3}$

c. $\frac{\sqrt{3}}{2}$

d. $\frac{4}{3}$

© ClipSAT — clipsat.org | All Rights Reserved